



# CS & IT ENGINEERING

## Discrete Mathematics

Set Theory & Algebra

DPP 03 Discussion notes

By- Mili Dhara Ma'am



[NAT]

GATE - 2024



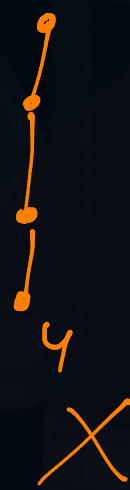
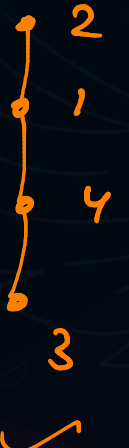
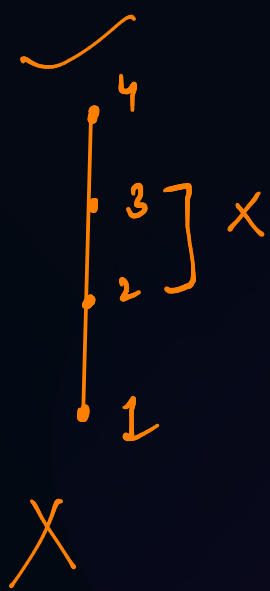
#Q. Let P be the partial order defined on the set {1,2,3,4} as follows

$$P = \{(x,x) \mid x \in \{1,2,3,4\}\} \cup \{(1,2), (3,2), (3,4)\}$$

The number of total orders on {1,2,3,4} that contains P is

4  
Total orders  
=  $4! = 24$

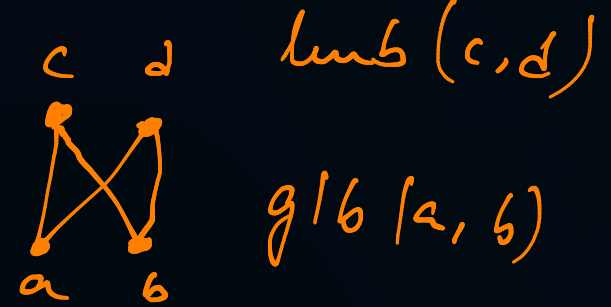
$$P = \{ \underbrace{(1,1), (2,2), (3,3), (4,4)}_{\downarrow}, \underbrace{(1,2), (3,2), (3,4)}_{\downarrow} \}$$



5

1, 2, 3

#Q. Which of the following is/are always true for any lattice?



Bounded lattice  $(N, \leq)$

~~A~~

There exists exactly one minimum and exactly one maximum element

~~B~~

There exists at most one minimal and at most one maximal element

~~C~~

Least upper bound and greatest lower bound exists for every pair of elements.

~~D~~

Every element has a unique complement.



#Q. Let  $f : A \rightarrow B$  be a function, and let  $E$  and  $F$  be subsets of  $A$ . Consider the following statements about images.

S1:  $f(E \cup F) = f(E) \cup f(F)$

S2:  $f(E \cap F) \subseteq f(E) \cap f(F)$

S3:  $f(E \cap F) = f(E) \cap f(F)$  if function  $f$  is one-one

Which of the following is true?

**A**

Only S2 and S3 are true.

**B**

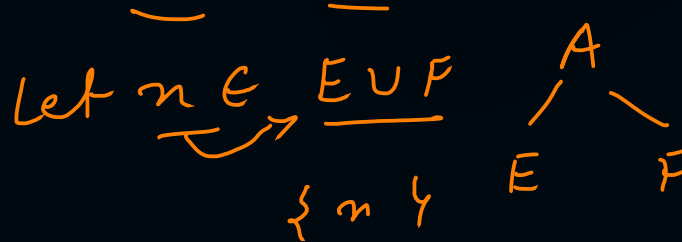
Only S1 and S3 are true.

**C**

Only S1 and S2 are true.

**D**

All S1, S2 and S3 are true.



$$f(x) = y$$

$$x \in E$$

$$f(x) = y$$

$$x, y \rightarrow E \cap F$$

$$F \subset P$$

$$f(p) = 9$$

$$f(q) = 6$$

#Q. Let  $f(x) = \frac{x}{x+1}$  and  $g(x) = \frac{x}{1-x}$  then  $(f \circ g)^{-1}(x) = ?$

$$f \circ g = f(g(x))$$

$$\begin{matrix} x \\ x \end{matrix}$$

$$y = f\left(\frac{x}{1-x}\right)$$

$$y = \frac{\frac{x}{1-x}}{\frac{x}{1-x} + 1} = \frac{\frac{x}{1-x}}{\frac{x + 1 - x}{1-x}} = \frac{x}{1-x} \times \frac{1-x}{1} = x$$

#Q. Let  $A = \mathbb{R} - \{3\}$

$B = \mathbb{R} - \{1\}$

Where  $\mathbb{R}$  is the set of all real numbers if

$f: A \rightarrow B$  such that  $f(x) = \frac{x-2}{x-3}$ , then ?

**A**

$f$  is one-one but not onto

**B**

$f$  is not one-one but onto

**C**

$f$  is bijection

**D**

$f$  is neither one-one nor onto

$p, q$

$$\frac{p-2}{p-3} = \frac{q-2}{q-3} \rightarrow p=q$$

$p=q$

$$\Rightarrow pq - 3p - 2q + 6$$

$$= pq - 2p - 3q + 6$$

$$\Rightarrow q = p$$

$$y = \frac{x-2}{x-3}$$

$$\Rightarrow yx - 3y = x - 2$$

$$yx - x = 3y - 2$$

$$x(y-1) = 3y-2 \Rightarrow x = \frac{3y-2}{y-1}$$

#Q. If the function  $f: [1, \infty) \rightarrow [1, \infty)$  defined by  $f(x) = 2^{x(x-1)}$  is invertible, then  $f^{-1}(x)$  is ?

**A**  $\frac{1 + \sqrt{1 + 2 \log_2 x}}{4}$

**B**  $\frac{1 - \sqrt{1 + 2 \log_2 x}}{4}$

**C**  $\frac{1 + \sqrt{1 + 4 \log_2 x}}{2}$  ✓

**D**  $\frac{1 - \sqrt{1 + 4 \log_2 x}}{2}$

$$y = 2^{x(x-1)}$$

$$\Rightarrow \log y = x(x-1)$$

$$\Rightarrow 0 = x^2 - x - \log y$$

$$x = \frac{1 \pm \sqrt{1 + 4 \log(y)}}{2}$$

# [MCQ]



#Q. The domain of the function

$$f(x) = \sin \left\{ \log \left( \frac{\sqrt{4-x^2}}{(1-x)} \right) \right\} \text{ is?}$$

$\downarrow$   
+ve

$N < 0$

$D < 0$

$$4-x^2 < 0 \quad \times$$

$$N > 0$$

$$D > 0$$

$$\sqrt{4-x^2} > 0$$

$$1-x > 0$$

$$\Rightarrow 4-x^2 > 0$$

$$\Rightarrow 1 > x$$

$$\Rightarrow -2 < x < 2$$

$$\underline{-2 < x < 1}$$

**A**  $(-2, 0)$

**B**  $(-2, 1)$

**C**  $(-2, 2) - \{1\}$

**D**  $(0, 2) - \{1\}$



#Q. Which of the following statements is /are True ?

**A**

A constant function is one-one iff domain of the functions has exactly one element

**B**

A constant function is onto iff co-domain of the functions has exactly one element

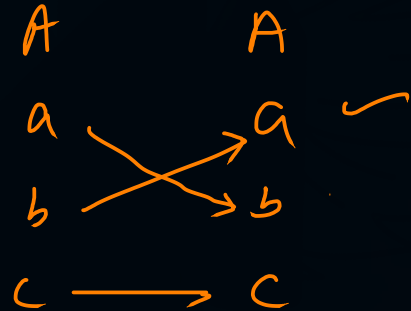
**C**

Every one-one function on a finite set A is bijection

**D**

Every onto function on a finite set A is bijection.

$$f(x) = k$$

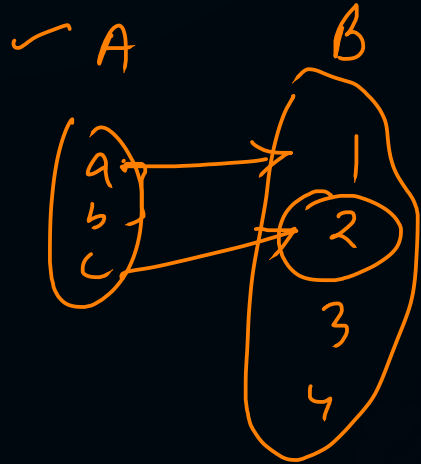


#Q. Let  $A = \{a, b, c\}$  and  $B = \{1, 2, 3, 4\}$  Set C is defined as

$$C = \{f : A \rightarrow B \mid 2 \in f(x) \text{ for some } x \in A, \text{ and } f \text{ is not one-one}\}$$

Then number of elements in set C are ?

functions



Case I image is 2 for one element of A  
 ${}^3C_1 - 3 \times 3 = 9$

a — 3 —  
 b x

Case II: Two elements

$${}^3C_2 = 3 \times 3 = 9$$

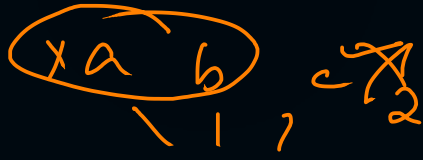
Case III

— 1 way

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#Q. Let  $|A|=6$  and  $|B|=3$ , then number of onto functions possible from set A to set B are \_\_\_\_\_?



Total fun -  $3^6 = 729$

$1^6 = 1$

$$\begin{aligned} |S_a \cup S_b \cup S_c| &= |S_a| + |S_b| + |S_c| - |S_a \cap S_b| - |S_a \cap S_c| - |S_b \cap S_c| + |S_a \cap S_b \cap S_c| \\ &= 64 + 64 + 64 - 1 - 1 - 1 + 0 \\ &= 189 \end{aligned}$$

Onto =  $729 - 189 = 540$



**THANK - YOU**